FREE

Statins Are Mitochondrial Toxins

And CoQ10 isn't enough to fix this.

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Blood cholesterol levels, along with LDL particle count, ApoB concentration, LDL particle size, LDL pattern, and the ratio of cholesterol between LDL and HDL are all governed centrally by the LDL receptor.

Genetic mutations in LDLR cause familial hypercholesterolemia by lowering LDL receptor expression. One in a million people are born with homozygous mutations, and this can cause heart attacks as young as 18 months old. One in 300 people are born with heterozygous mutations, which shifts heart disease forward from a phenomenon that primarily kills between the age of 60 to 80 to one that primarily kills between the ages of 35 and 60.

The LDL receptor is the main way you bring cholesterol from outside the cell to inside the cell. While the liver only makes about 16% of the body's cholesterol, it is overwhelmingly responsible for controlling the concentration of cholesterol in the blood by taking it up using the LDL receptor. That cholesterol can then be put to productive use, especially for the synthesis of bile acids to support digestion.

However, there are two primary governors of LDL receptor production:

- **Thyroid hormone** governs it by signaling **abundance.** This causes the liver to take up *more* cholesterol than it needs to meet its basic needs, and *this* is what drives bile acid production to support digestion.
- **Cholesterol deficiency** within the liver cell governs it by signaling **scarcity**. The liver takes up cholesterol from the blood not to put it to productive use but simply to replenish its own stores.

Statins work by inducing a cholesterol deficiency in the liver. The liver responds to this by taking up more from the blood to bring its stores back up to normal.

In his 1976 book, *Solved: The Riddle of Heart Attacks*, Broda Barnes reviewed the history of using thyroid hormone to control both cholesterol levels and heart disease risk. It worked extremely well, but it was discontinued due to what Barnes describes as irresponsible dosing that had killed a few people.

It is important to understand here that you do **not** need to be “hypothyroid” for thyroid hormone to lower your cholesterol. It will work at any dose. Practitioners using it did not realize they were **playing with fire** by **using a hormone to signal a degree of abundance that was not present in the body.** Thus, some of them let the dosing get out of hand because it had nothing necessarily to do with correcting a thyroid deficiency.

Nevertheless, their approach was the same as using statins because **both thyroid hormone and statins increase the LDL receptor.**

On the other hand, it was the opposite, because **thyroid signals abundance while statins signal scarcity.**

What everyone involved fails to fundamentally understand is that **your LDL receptor does not work in a vacuum. It is powered by mitochondrial ATP production**.

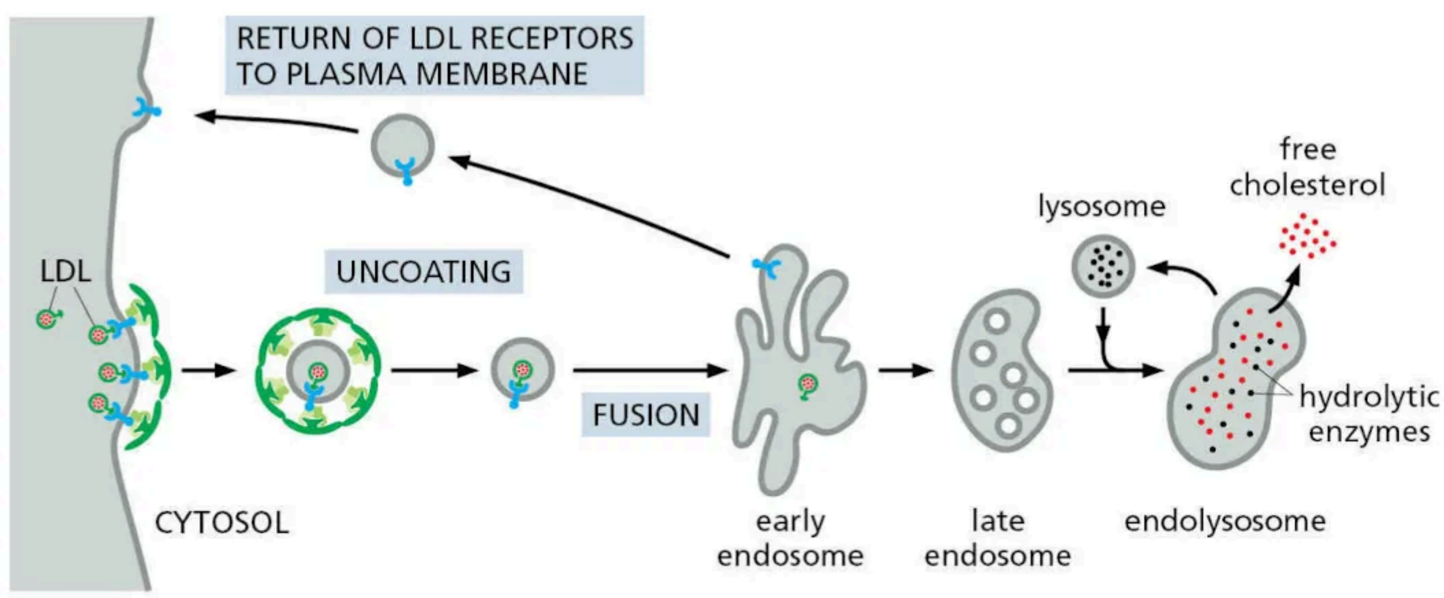
If your mitochondria are not working well, your LDL receptor simply will not work.

Doctors who use a pharma-first approach have little respect for nature and doctors who use a statin-first, mitochondria-never approach do so because they do not understand the most basic elements of cellular biology.

As covered in What Everyone Should Be Doing For Their Health, everyone must use a food-first, pharma-last approach, which means everyone needs their doctor to follow the same approach. Do your research until you find a good one.

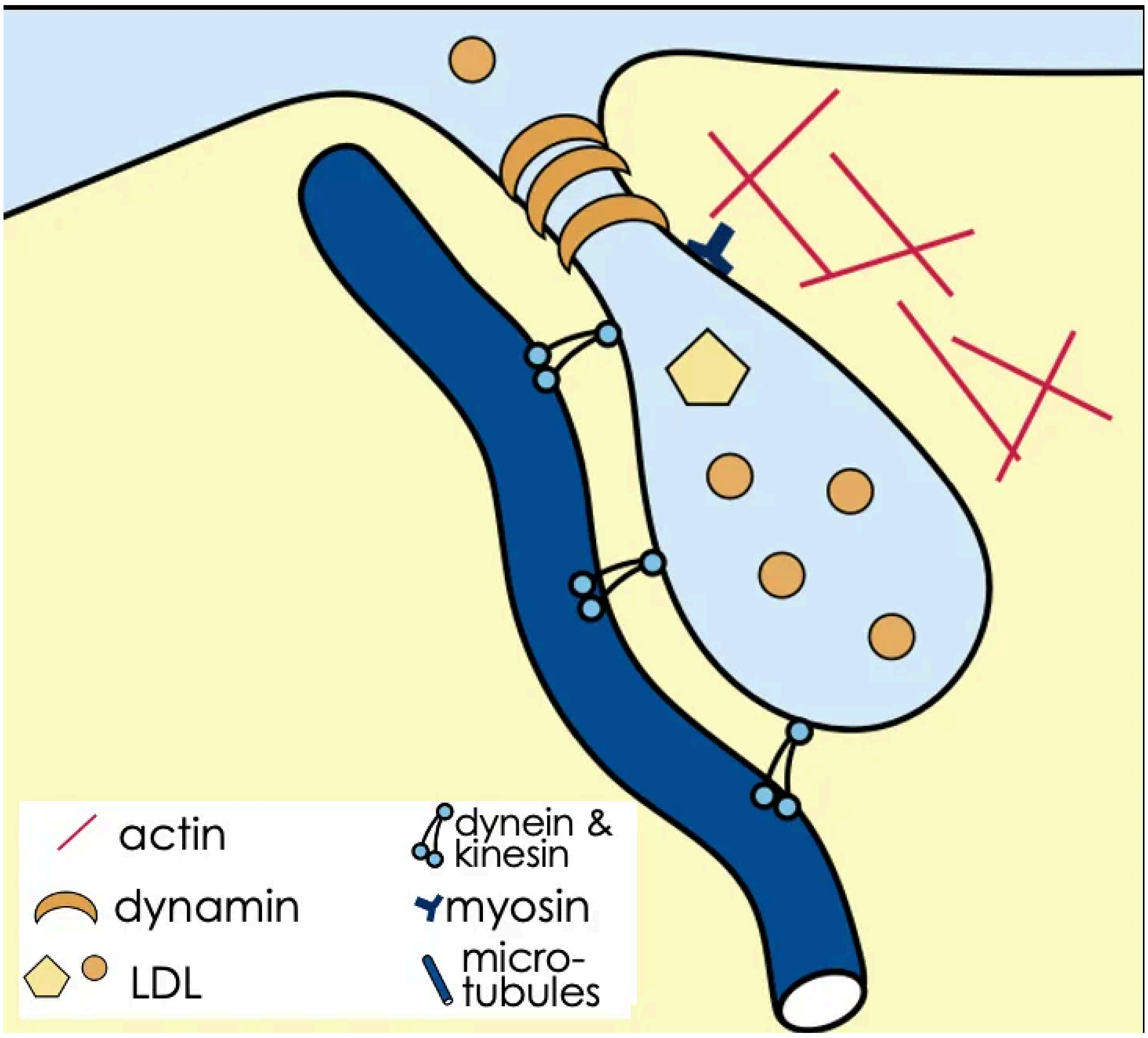
How Mitochondria Power Your LDL Receptor

As shown in this diagram from Alberts, *Molecular Biology of the Cell*, the "Bible" of molecular and cellular biology, the LDL receptor clears cholesterol from the blood by binding to LDL particles on the cell surface.



They are present in the membrane bound to the green protein, clathrin, which helps facilitate the membrane invaginating and pinching off to form a vesicle. The protein coat must then be removed, and the vesicle fuses with an organelle known as the endosome, which then uses the digestive organelle down as the lysosome to eat up the contents, digest them, and release the cholesterol.

The figure below shows more detail:



The protein **dynamin** squeezes the invagination to pinch it off from the cell membrane.

Have you ever tried to *squeeze* anything without investing *energy* into it? Here's something you could try: buy a dynamometer and see what your grip strength is when you feel most awake and energetic. Then try it just before you go to bed, immediately upon waking, or after you finish a grueling workout.

No energy? No squeeze.

Dynamin is powered by the hydrolysis of GTP, ATP's close cousin. In fact, GTP production is directly fueled by ATP, so without mitochondrial ATP production dynamin won't function.

Put all the LDL receptors in your membrane you want. If the vesicle can't squeeze off, the cholesterol is staying in your blood.

Contrary to what some of us learned in high school biology, cells are not sacs of fluid. Things don't come in and then just float around. They are highly organized microcosms of civilization, with several classes of proteins making up their structural infrastructure and transport superhighways known as the "cytoskeleton."

The two major cytoskeletal proteins involved in LDL uptake are actin (red) and microtubules (blue).

Actin filaments organize themselves around the invagination to provide structural support to it. They interact with it using **myosin VI.** This protein is in the same family as the myosin II that performs skeletal muscle contraction. It is **directly powered by ATP.**

The microtubules make up the transport superhighway network within the cell that, in this case, facilitates the newly formed vesicle moving toward the endosome. The motor proteins that power movement along the microtubules are **dynein** and various **kinesins**, which are **directly powered by ATP.**

Not shown in the figure, the **uncoating** of the vesicle is performed by **heat-shock protein 70** (HSP70), which, like every other heat-shock protein, is **directly powered by ATP.**

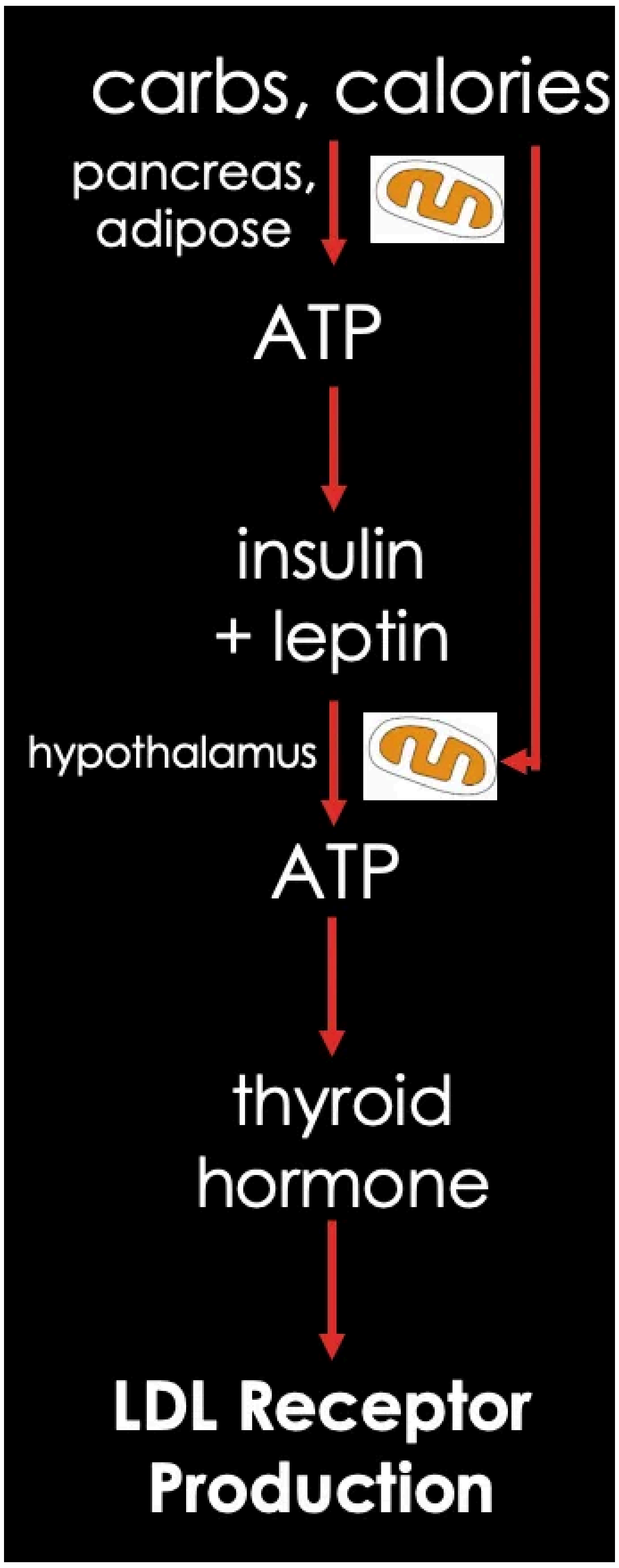
Thus, every LDL receptor that brings an LDL particle in from your blood requires five different classes of ATP-dependent protein to actually move it from the blood into the cell.

No mitochondrial ATP?

No LDL receptor activity.

How Mitochondria Power Your Thyroid Hormone

As I covered in Hormones Are Never In Charge, thyroid hormone is primarily regulated by mitochondrial ATP production.



Caloric intake, especially carbohydrate, drives the pancreas to make insulin and the adipose tissue to make leptin. Adipose production of leptin is at baseline proportional to the total fat stored, so insulin is a gauge of short-term energy status and leptin is primarily a gauge of long-term energy status.

However, these hormones are not directly stimulated by carbs and calories. They are stimulated instead by **mitochondrial conversion of food to ATP.**

Insulin and leptin both act on the hypothalamus to increase thyroid hormone-releasing hormone (TRH). They do so by increasing ATP production in the hypothalamus.

Insulin and TRH both act together on the thyroid gland to increase the output of thyroid hormone.

Thus, any impairment in mitochondrial ATP production will hurt thyroid hormone production.

Conversely, **efficient mitochondrial conversion of food to ATP will drive thyroid hormone secretion** because it serves as a **highly accurate signal of true energetic** abundance.

Optimal Mitochondrial Function Optimizes Blood Lipids

Thus, optimal mitochondrial ATP production exerts a concerted effect on thyroid hormone to increase the LDL receptor and on the five ATP-dependent motor proteins that help bring the LDL particles into the cell.

What happens when you intervene to pump up the LDL receptor without creating the *actual abundance* necessary for its proper function?

On the one hand, it will not work as well as it would if ATP were abundant.

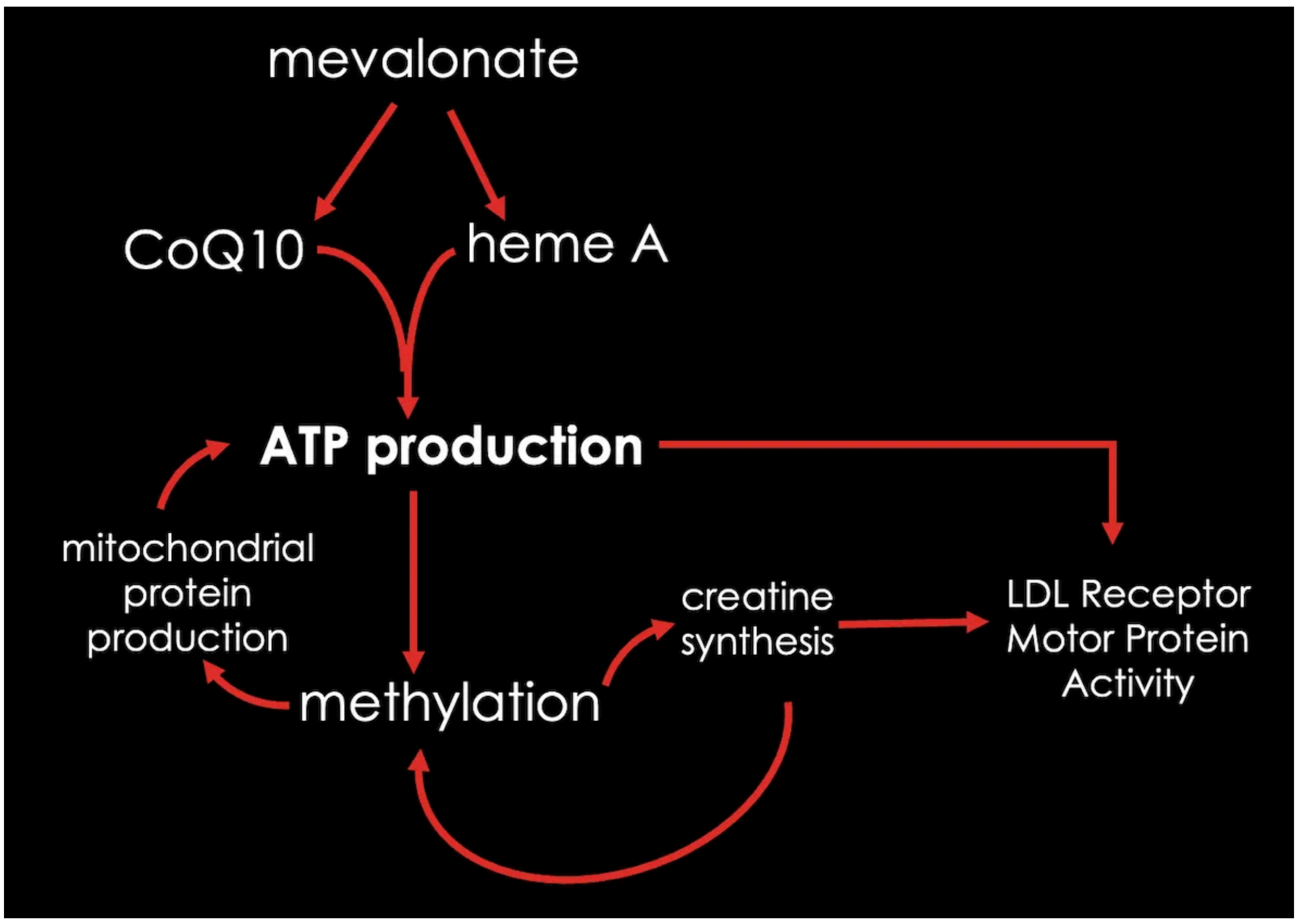
On the other hand, more LDL receptors will try to use up more ATP, so they will take it from what is available, which will make less available for other things.

Such a solution is like seeing an oil shortage and solving it by importing more cars. Will you incentivize more driving? Yes, you will. Most likely, more people will get from A to B. But now you've driven up home heating prices for people who heat their homes with oil, so now you'll have more people with hypothermia, hypertension, cardiovascular and respiratory problems, and misery.

In other words, you are definitely causing cholesterol to move from your blood to your cells. But at what cost? You are draining your ATP supply and problems are just going to pop up somewhere else.

Statins Are Mitochondrial Toxins

Statins induce cholesterol deficiency in liver cells by impairing the production of mevalonate, a compound far upstream from cholesterol.



Mevalonate is used to produce *many* different compounds, but two that are central to mitochondrial function are CoQ10 and heme A.

Heme A is the specific iron-based protein that facilitates the use of the oxygen to convert food to ATP in the mitochondria.

By decreasing ATP production, statins hurt methylation. This is because ATP is used to make methylfolate and to activate methionine to S-adenosyl-methionine, the universal methyl donor.

Methylation is needed to produce [all mitochondrial proteins](). Since methylation is needed to make ATP and ATP is needed for methylation, a decline in methylation that feeds back on mitochondrial ATP production will initiate a vicious cycle.

Methylation is needed to produce creatine, and this is probably why [statins decrease creatine synthesis]().

Creatine is the mitochondrial power grid. It spreads the impact of mitochondrial ATP production throughout the cell, especially in the general area of the cell known as the cytosol.

It is in the cytosol where S-adenosyl-methionine is made, so creatine supports its production and a decline in creatine status will compromise methylation in a vicious cycle.

Thus, by the time low ATP is hurting methylation beyond a certain threshold, you've locked in a **vicious supercycle** composed of multiple vicious cycles that will cause perpetually declining mitochondrial function.

You now have less ATP and less creatine, which are two hits *against* LDL receptor activity, even though you have increased the production of that receptor.

To take this back to our oil analogy, this is worse than importing cars to deal with an oil shortage. Rather, this is *denying drilling permits* so that the supply of oil gets cut in half, then *diverting half the oil* left over into importing those cars. Now you increase demand after you've already decreased supply. You still get more people getting from A to B, but your heating oil shortage has gotten a lot worse and you should probably prepare for a popular revolt among the poor whose children are frostbit and whose grandparents are dying.

In other words, you are hurting the total supply of ATP but diverting what ATP you have toward LDL receptor function. It works to lower cholesterol, but at the cost of other processes that need cellular energy, which is… all of them.

CoQ10 Is Not Sufficient to Prevent or Reverse Statin Myopathy

Statins can cause muscle pain and cramps that can get bad enough in a small number of people to cause exercise intolerance and even rhabdomyolysis. "Rhabdo" is a result of muscle damage that causes muscle proteins to turn the urine a dark color.

As reviewed here, clinical trials only show this happening in 1.5-3% of people, whereas observational studies show that it occurs in 7-38% of people.

The discrepancy can be explained by the fact that people who sign up for clinical trials are healthier than the general population, the trials screen for vulnerability to side effects far more strictly than doctors do for their patients, and you can control the incidence by how strictly you define myopathy. For example, if you count people who have an increase in unexplained cramping, you will get a large number, and if you only count people whose rhabdo put them in the ER you will get a small number.

Meta-analyses conflict as to whether CoQ10 does absolutely nothing or helps substantially with statin-induced myopathy. For example, a 2022 meta-analysis concluded CoQ10 does nothing, and a 2025 meta-analysis concluded that 100-600 milligrams per day has a very helpful effect.

Although the 2025 analysis is more recent, only a single 2022 trial contributing to 16% of the result was published after the 2022 meta-analysis was prepared. The primary difference between the two is that the 2022 meta-analysis required creatine kinase data to verify the myopathy and only included placebo-controlled trials whereas the 2025 meta-analysis required only a survey of muscle pain and included trials that were not placebo-controlled.

Regardless, even if we treat the 2025 analysis charitably, it indicates that **CoQ10 is only partially effective**. For example, if the subjects reported a muscle pain score of 5-6, CoQ10 would reduce this by 2.5 points, which indicates **it could roughly cut the muscle pain in half**.

Cutting it in half is good, but if that's where the result ends, it isn't a resolution. Moreover, if the **vicious supercycle of declining mitochondrial function** has not been reversed, it is likely to keep getting worse after the trial is over.

Creatine supplementation has led to the complete resolution of statin myopathy in some cases, but has only been studied in 13 people, published in a case series and a case report.

Metabolomic analysis of statin-associated myopathy cases shows that there are broad shifts in energy metabolism that cannot be attributed to any one single cause and are idiosyncratic between patients. For example, there is a broad shift away from branched-chain amino acid metabolism and toward arginine and proline metabolism, which I suspect reflects impairment in the conversion of pantothenic acid (vitamin B5) to the energy carrier coenzyme A (CoA) as a secondary side effect of the **vicious supercycle**. There is a broad shift toward altered glycine and serine metabolism, which I suspect reflects **impaired methylation**. Some people have increased carnitine synthesis and others have decreased carnitine synthesis, showing that the **mitochondrial dysfunction is idiosyncratic**.

It is possible that some interventions like CoQ10 and creatine can in some people break one or two of the vicious cycles enough to interrupt the vicious supercycle, but neither of these are going to normalize the production of heme A.

There is no supplement that can reverse the inhibition of heme A synthesis, which means nothing can resolve the fact that **statins will impair your mitochondrial utilization of oxygen to produce ATP.**

Mitome Can Examine What Is Unique to You

As the Founder and Scientific Director of Mitome, I embedded in the reporting the ability to differentiate between impairments in CoQ10-mediated processes, signatures that methylation deficiency is impairing mitochondrial function, and the impact of inadequate complex IV activity that would be caused by a disruption in heme A.

Mitome directly tests the respiratory chain components, where disruptions of CoQ10 and heme A synthesis would first be found, and thus looks at the root cause of the mitochondrial vicious supercycle.

If removal of statins and supplementation with CoQ10 and creatine do not resolve statin myopathy, Mitome could determine what is still missing and what to do about it.

Mitome is not meant to diagnose or treat disease and has not been clinically tested in statin-associated myopathy.

However, it can reveal what the unique limitation in your mitochondrial function is and provide you with an actionable protocol to address it with food and supplements.

Why Mitochondria Must Come First

If you optimize your mitochondrial function, you promote robust thyroid hormone activity to increase your LDL receptor production, and you promote robust supply of mitochondrial ATP production to make the five motor proteins needed for LDL receptor activity do their job.

You also need to be eating enough protein, iodine, and antioxidant nutrients to support your thyroid.

To examine your diet, use the guidelines in What Everyone Should Be Doing For Their Health, track your micronutrients in Cronometer, and run the Comprehensive Nutritional Screening.

If your cholesterol or LDL-C or ApoB or whatever you are measuring remains concerningly high despite adequate micronutrients, and you have also followed the guidelines in What Everyone Should Be Doing For Their Health regarding body composition and stress management, you should experiment with fiber, your fat-to-carbohydrate ratio, and your intake of saturated fat and cholesterol.

In general you need to get your BMI to 18-25 or more precisely optimize your body fat, and, in that context, push your carbohydrates up to a point that does not destabilize your glucose and lactate numbers or make you regain body fat.

More fiber will, in general, help lower cholesterol, but your fiber intake should also be adjusted to what promotes optimal digestion and stool function.

If you have chronic inflammation that is not attributable to excess body fat, you can try a short 12-week course of any of the following, alone or combined: 3 capsules of cod liver oil and 6 capsules of fish oil for high-dose omega-3s; boswellia; 250-2500 milligrams of Specialized Pro-Resolving Mediators; 1000 milligrams once or twice per day of black seed oil; or 1.2 grams (3 capsules) of curcumin as Biocurcumax; or the lactoferrin protocol.

If these do not resolve your blood lipid problem you can try restricting saturated fat, focused on the foods that provide you with the least important micronutrients. You can compare this to restricting cholesterol, again focused on restricting the foods with the least important micronutrients. You can combine these if they both seem to work. While replacing saturated fat with polyunsaturated fat will work, I do not consider it wise.

If after all this you still need to work on your blood lipids you could consider a statin. It may in fact be optimal in certain extreme cases.

However, you should always use a food-first, pharma-last approach to everything, and you should always use a mitochondria-first, statin-last approach to blood lipids.

If you optimize your mitochondria first and you still need more LDL receptor activity, then by the time you intervene, the LDL receptor will be using the most abundant supply of ATP you were able to provide it with.

If instead you use statins — unambiguous mitochondrial toxins — as your *first* approach, you are killing the ATP supply and then making the LDL receptor take what little there is from the rest of your functions that need it.

It is unlikely that even the most rose-colored clinical trials and case reports are telling the full story. On average, you lose 1% of your mitochondrial function per year. While 100-600 milligrams of CoQ10 or 5 grams of creatine per day may be very helpful in statin-associated myopathy, will that be true ten years after the study was completed when mitochondrial function has another decade worth of decline under its belt? I doubt it.

Age only explains 25% of mitochondrial function, which means that 75% is in your power to optimize.

Focus on getting incredible mitochondria first so you don't have to try figuring out the cause of mitochondrial failure later.

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Genjaji 39m

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1 reply by Chris Masterjohn, PhD

Fernán 1h

Hey Chris, thank you so much for what you are doing. The big missing piece is how SSRIs affect the gut/liver, and how to help revert and support the body during and after tapering. Many people including myself report that the gut isn't the same but can't piece exactly why.

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3 replies by Chris Masterjohn, PhD and others

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